

ActInf GuestStream 060.1 ~ "Reflections and implications of a naïve attitude"

GuestStream | 2:06:31

Hello and welcome to the Active Inference Institute.

This is ActInf Guest Stream number 60.1 on October 18th, 2023.

Today should be a great and very exciting presentation and discussion.

Kristen Whittle is here and will be presenting on reflections and implications of a naïve attitude to architecture.

There will be a presentation followed by a discussion.

So if you're watching live, please feel free to write any comments in the chat.

Otherwise, thank you so much for joining and really looking forward to this work being shared.

Thank you, Daniel.

It's a great pleasure being able to present to you and your audience.

And just as we were maybe talking, the journey towards arriving here to talk to you has been a very, very interesting one and one that's really been undertaken since I was a young person.

So it's thrilling and exciting to be able to talk about the work that we've been doing as a studio and kind of seeing where the touch points and points of influence and interest to you and your audience come into this conversation about our sort of artistic work, I guess.

So what I wanted to do is to maybe explain what we're just about to see before showing it.

What it is, is a kind of an assemblage of work that's been undertaken since I was probably 22.

And it goes into a lot of practical work that has now been completed, real large scale buildings, etc.

So it's kind of part theoretical part, kind of trying to understand as a curious person where I'm at and trying to evolve a maybe a strong systematic view on architecture, because I guess the way I see it is that architecture is a bit lost.

And in a sense, there's an idea implicit in the work, which is to kind of remind ourselves the purpose of architecture in a kind of pre-industrial form,

and then looking towards the future, towards a very sophisticated future that we have in front of us and trying to synchronize that with a very humanist substrate, humanist philosophy.

So the reason why I called the talk a kind of evocation about naïve architecture actually comes back to some readings that we've been doing,

and that Carl Jung evolved out of readings that he'd had from Schiller, the German writer and philosopher about ways in which artistic production occur.

So one would have, for instance, one production of art stemming from a natural source almost, which he referred to as naïve art, or a naïve approach to art.

And the other, which is what he referred to as a sentimental view, or a sentimental approach, which is more that you're standing away from nature and looking upon it and choosing which direction to evolve in and objectifying the direction.

So the first thing to say is that our work really stems from an extrapolation of nature through our work in a kind of intuitive, instinctive manner.

So somehow within this, there's a correlation between how cognition works, which I see as the foundation to creativity, and then how the making process works associated with that and perfecting the making process. And then obviously, what the lived experience of those environments is,

how does that feedback work to go back into cognition and creativity? So I guess some references here might be useful. The work that Smuts did, you know, at the same time as when Jung was working in Basel, trying to evolve a complex holistic approach to the mind and body has been instrumental in the work that we do.

So in a sense, one takes that sort of expansive, infinitely expansive kind of perspective of the universe being fed into us, and then us emitting that universe, that sort of overall large scale complex process that's always happening,

is somehow our creative process is captured within that. So I use that as a bit of a reference point.

And I'd like to just, I guess, put a shout out here to Corbusier, because even though he manifests the scientific,

and maybe definitely institutional, and scientific approach to the modular human modular,

his focus was really on the body. His artistic, maybe intuitive side, was maybe interested in the mind,

but the systems and modernistic rationale that he produced in the 20th century had a massive effect on cities globally,

and ultimately kind of fell over post-brutalism into a kind of failure of architecture.

I see the work that we're doing is somehow an evolution of the modular,

and a much more complex and diverse manifestation of the modular.

So I guess coming back to Carl Jung again, it would be necessary for me to say that

if you go along with the story of naivety and processing nature, that one of those processes is to do with the archetypal image,

and the sort of using the archetypal image as some sort of foundation for the work.

So if this sort of stream of consciousness is available to you, that's one of the latching points of the work.

And the question then becomes, how do you utilize that as some sort of febrile substance to manifest new things?

So not necessarily reproducing memory as an exact thing in and of itself.

It's more associated with the creative utility of the archetype as a substrate to tap into,

and then applying different processes onto that.

And because of that, let's say, use of that substrate,

we're able to get into a kind of deep relationship or a deep conversation with the public.

So I think the reason why this is kind of important at this moment in time is, you know, the question about data.

What data informs architecture?

Typically, it's been ergonomic fact-based information that's determined by spatial mathematics, and ergonomic programming of space, like utility, efficiency, and those sorts of concepts.

I'm sort of more interested in the pre-industrial, which is more to do with scale,

the idea of rooms, the idea of enclosure and connection to other people,

and what sort of spatial configurations allow for that.

So the question that I have is really, how do we use data in that sense, you know,

outside of the engineering-based, mathematical, scientific, modernistic platform for data?

So which I think causes institutionalisation, which has been seen as a failure in architecture and failure in society.

So we're sort of, we've never recovered really from the failure of the institutionalisation of architecture. So we're now at the point with large data and AI to question what gets assembled from here on in, like, and how do we evolve something that's got the nuanced complexity freedom of our organic cells in a kind of artificialised system?

So I'm very interested in individuation, both as a creative person, but also creating environments that allow for other people to be individuated and for allowing that ecosystem to dominate.

So I'm therefore interested in industrial experimentation or experimentalism and how mass customisation can occur using new manufacturing techniques like, you know, additive manufacturing, for instance, and in a sense, democratising the design process to allow everyone to be an artist, to be an architect.

But anyway, so to kind of unpack what we would do typically is rather than go on a linear journey, which is a fact-based, reduced, simplistic architectural journey, like $A + B = C$, we go on a circular kind of coiling journey.

It's certainly moving forward, but information's coming in from different vantage points continually throughout the process.

And I guess we've ended up being very efficient at this, but it allows us to have kind of oppositional trajectories coming into the design journey. It allows us to see things objectively as well as subjectively, using qualitative and quantitative information, and building things and destroying things on a journey of creation.

So things die, certain things live, the living things get tested and might die, and it evolves in that sort of way.

Okay, so if you're looking at maybe the definitions of creativity, it would probably centre on how creative processes end up creating new things, it's like not, so to speak, about mimicking.

It's about creating new things that have a function.

And I guess from our perspective, this coiling process is the foundation to that invention process, where somehow you yourself, because you yourself are connected into that information process, that by necessity what comes out of it traps the external world into that internal world and gets affected by that journey.

So it's sort of a trap for society.

But the main objective is for one to achieve what I would call coherence, which definitely comes back to free energy and active inference, where you can essentially map and see the world moving forward in a coherent way, latch onto it and for it to feedback meaning and reward into your life

and therefore extend longevity and improve livability, those sorts of things.

So by sort of going in that direction,

what we need to do is to kind of assess what the well of knowledge plus memory plus ambition, what sort of constitutes that domain.

So what we've done in our own way is to assemble trajectories

and a kind of taxonomy, a working taxonomy that ranges between social and political principles to scientific and technological principles and art and architectural principles.

And that essentially forms a kind of rosette, which we call deep learning.

And that's the kind of foundation that has existed for me since childhood and obviously is a dynamic assembly process.

So that's the kind of scaffold.

And then if you imagine taking that living scaffold to a project,

we then create a dynamic relationship between that scaffold to a client

and the projections and information that we get to and from a client,

as well as to and from a site.

So a site would, for instance, constitute where the sun moves,

how it moves around a site, where the wind blows, what's the ecology of a site.

And in a city, what are its neighbours?

What's the social backdrop to a site?

And it's also about people, obviously.

And it's about monitoring and assessing people

and how they are coming and going within a site's curtilage.

So you're sort of assembling that very broad and complex picture

that essentially forms that nugget of truth, which we've defined as feeling.

And so we trust our feelings about a site.

Now, it sounds naive talking about feelings in architecture.

And there's no sort of, I don't believe there's any sort of academic construct that's been developed around feelings.

It's been sort of chipped away in rationalist principles in postmodernist theory.

But the idea of a feeling is obviously if you're trusting your instincts,

it's a very complex model of truth that's a truth to you.

And so if you're able to form a layered understanding

about how that feeling is constructed,

you're then able to, I guess, talk about it openly with your studio

and with your clients and start utilising it and testing it more pragmatically.

So what we've done is to then layer that out from initiation phases

all the way through schematic delivery and post-occupancy phases of projects.

So that feeling concept is the nugget that everything is tested against.

And so when we get to the end of the process,
we're measuring against what that original feeling was and asking questions,
does it feel like what we felt like at the beginning?
How does it differ?
And what do other people think about that feeling?
And I guess it's probably important for me to say
the curiosity that underpins all of this
started because the work that was happening
ended up being very well fitted to the client's prerogatives.
And I would have a situation quite commonly
where a client would say to me,
how did you read my mind?
Like, how were you able to comprehend what our needs were
when there was a lot of unspoken truth about a project?
How were you able to encapsulate what our desires were in a project?
How is that possible?
And or in the sense of in the hospital,
why does the hospital feel right?
Why does it feel good?
Why is it restorative?
Like, how have you been able to achieve that?
And because of these questions that came at me,
I decided back in 2019 to study it,
to work out why it was happening,
what fed it, what underpinned it.
So that's kind of what why I'm here really talking to you, Daniel.
And I guess what I've realized is that
that the perception model
that I've just talked about in terms of feeling
has obviously a corollary or an association with memory.
and anticipation.
So projection.
And somehow,
in a very complex neurological process,
we're capturing that data
and then we're able to project it into architectural models.
So if you imagine cognition,
like intuitive cognition projecting scenarios

to help guide your body through space.

What I do as an architect is I actually make models
of that projection.

And I then test those models
in order to get the perfect model,
the thing that's the most coherent model for me
and for the situation.

So it's a situationally specific coherent model.

That's what I'm searching for.

And that's what I build at the end of the day.

So it's based on feedback.

So we would call it an embodied holistic code
for architecture.

Okay.

So it's based on the idea
that we are all modular people
in the sense that the foundation for us all,
our neurobiological foundations,
are common
before they become idiosyncratic,
so to speak.

I'm sure you'll correct me on all that.

So anyway,
but it's fair to say
that we are all living beings, right?

And we have all of these
normal, instinctive,
kind of primitively derived
things that we do in life.

And this is what unifies us.

So as an architect,
it's important for me to know
how these actions work
work because they actually underpin
how the functioning capacity
for building works.

But it's also true to say
that in the work,

we are trying to reduce anxiety,
not invoke anxiety.

We're trying to create harmony
rather than invoke complexity
to the point where it creates
pathologies.

We're trying to harmonize our existence
in cities specifically.

And in contemporary cities,
this is where we go drastically wrong
because the human being
is never at the center
of the conversation
with regards to a development.

It's always a spreadsheet-driven exercise.

And this is where we go wrong.

So our work begins with studying
how this works,
how this kind of human environment works.

These are Carsten Holler's art piece,
this kind of objectifying
the social domain of a restaurant,
which is very evocative.

But it's a basic process.

And this is, again,
humble me with an iPhone.

What I try to do all the time,
every day,

is to document

what I would call nooks

where people are settling

and where people are seeking

to settle in certain domains,

whether that's to socialize

or to have the sort of intimate moments
of self-reflection.

How do we move through cities?

So what I'm always doing

is photographing
basically cities and people.
And that forms the foundation.
So I call that,
and again,
this is a very unacademic terminology,
learn, dwell, nooks.
So I've worked out that
my fascination
with photographing these things
stems from the fact
that I am learning something
from what I'm looking at.
And that's why I'm drawn to them.
And then also,
it's conflated with the idea
that I want to live
in that environment.
So there's something about
the domain
that I'm photographing
that combines
and conflates
learning something
and living in somewhere
that's going to be good for me.
So there's a conflation
of those two things.
So our work's very much
about the assemblage
and collecting
of that rationalized domain
of live, dwell, nooks.
So we're always fascinated
with how matter,
for instance,
assembles,
itself.

So this is an aerial photograph
of a region
of northern Victoria,
a desert region
of northern Victoria,
which defines this,
it's an irrigation colony
called Mildura,
which is a desert region
that became
an agricultural agrarian society
that was all to do
with growing food
for the city of Melbourne.
But if you look at it
from on top,
you get to see
the literal translation
of those ecosystems
being played out
by the patterns of nature.
So it's the patterns of nature
that we're interested in here.
We're interested in how
how we sanctify our existence
through symbolic constructs.
This again goes back
to the archetypes.
We're interested in materials
that are deep understanding
about the kind of exacting expression
of nature.
This is,
we have some fascinating trees
in Victoria,
in Australia,
and I became fascinated
in trees

when I did
the Royal Children's Hospital.
They're always
a foundation pillar for me.
Anyway,
materials that
may be a transformative materials
that are ambiguous,
especially drawn
to those sorts of materials.
materials that have
maybe a
handcrafted
and slightly sublime
complex process
that's created them.
Materials that
have used literal
tree trunks
in buildings
that maybe have
an unusually primitive
quality to them,
as well as
highly elaborate
expressions of roomness
and thresholds
that we see
in classical architecture
or humble,
quiet domains
that have
an atmosphere
of living dwelling.
There's a sort of
livingness
to an environment.
So I'm interested

in these subtleties
and then amplifying
those subtleties
in architecture.

The idea of
intimacy and privacy
in a public domain,
nooks of a doorway entry
or this,
I guess,
almost cinematic
memory
that's embedded
into cities.

The organic growth
of cities,
the bricolage
of cities,
how growth
happens
with different
personalities
and how they form
organic constructs
or basic
symbolic archetypal
classical city
terminology
like here
at pizza
in Italy.

So
and also
the organic
scaled,
mid-scale
cities
that you see

in Europe
that have
consolidated
that intimate
relationship
to space,
solid and void
and creating
spatial
constructs
where you have
external rooms
and internal
rooms
and the idea
of those spaces
aggregating
social activity.
So all of those
things are
overall fascinations
and I put them
in because they
are just
long-term
fascinations
which are
always being
recorded.
It's a kind
of like a
background noise.
But so the
origins of the
work go back
to PsyArc,
I think,
when I was

studying in
PsyArc
in Switzerland
in America
where I think
I was fascinated
by the idea
of coexisting
realities in
structures.
So I call this
the hyperrealism
of live
reality.
So this,
for instance,
is a sculptural
piece that I
did about
creating a
new construction
out of an
old construction.
So looking
at how an
old building's
column and
beam structure
was working
and how the
association of
the original
thought of that
structure,
which was used,
the drawings,
if you like,
the drawings

were the perfect
model of that
structure.

Then you have
the actualized
construction,
which is slightly
different than
the idealized
one.

And then you
have maybe the
course of time
and organic
processes changing
that structure.

And then, of
course, you have
your memory and
projection that's
pushed onto that
when you experience
it.

So this idea of
having an
assemblage of all
of those realities
made material
into a structure,
I became
fascinated by that
in order to
form a truer
basis for
architecture.

So that's kind
of like back
in the early

90s.

And then that
sort of morphed
into looking at
architecture more
like a body.

So looking at
the more
organic substrate
of architecture
being much more
similar to a body
and looking at
how a molding
process around
the body
can create a
replicable
architecture.

So a sort of,
in a sense,
a primitive version
of a data-fed
architectural model
being built
as a modular system
but replicating
more of a kind
of bodily expression.

So forming a very
intimate relationship
between body
architecture
and making a
very ambiguous
relationship there
rather than a
sort of

inside-outside
or dualistic
relationship.

This is much
more blurred.

So that,

I guess,

indicates maybe

an artistic

and curious

mind

being played

out as a

young architect

and then

fast-forward

that into

reality,

into professional

work.

I started to

work

back in

2007,

2008

in healthcare

projects

and this

was one

of my first

projects.

It was,

it's for the

most acute

care centre

for mental

health patients

in a kind

of lower
socioeconomic
district of
Melbourne
called
Dandenong
and what
we assembled
here is
an idea
of
de-institutionalising
a mental
health hospital
by using
the house
as the
model for
the building
and assembling
essentially those
like about 30
houses around
different courtyard
spaces and
making everything
out of
primarily recycled
timber
and then
changing the
relationship
between the
building
and the
people inside
but also
the people

inside the
building and
the public.

So de-institutionalising

all of those
relationships
and normalising

those sorts of
things where

people would
probably walk

past the

building and

think it was

a health spa

or some,

I don't know,

health hotel

but in fact

it's actually

treating the

most acutely

in need

patients inside

and then

using the

courtyards as

a kind of

natural oasis

to feed

positive energy

into those

people.

So these

buildings that

are closed

in and around

courtyard spaces

that focuses
the energy
towards the
courtyard
and each
courtyard
is designated
differently
and that
sort of
relationship
is more
like a
cloister.

So it's
more like
a kind
of
classical
cloister,
monastic
cloister
where peace
of mind
and mindfulness
comes about
through that
focus on
that singular
place of
peace.

So we're
using here
materiality,
nature that
somehow getting
in between
the building

and the
footpath
and then
the focus
of nature
on the
inside
to reassign
the relationships
between those
people's
minds
in and
of themselves
but also
their minds
in relation
to the
outside
world.

So these
are how
these sketches
and ideas
got translated
into spaces
that
changed
the
more hard-edged
relationship
of buildings
to nature,
private
ground
to public
realm
into much

more of a
harmonized
fluid,
fluid,
sequential,
picturesque
sequences of
space
where we're
able to
bring in
a more
what I would
call a
soft fascination
rather than a
hard fascination
reality for
those people
inside the
building.

So you can
see for
instance the
almost beach
like environment
around the
building that
forms the
border, the
framing of the
building before
you get to the
actual public
realm.

So it's a
kind of like an
imposition of

raw nature,
raw bush,
into the
public realm
that helps to
harmonize these
people.

And another
project that came
about slightly
later is a
very prestigious
physicist in
Melbourne who
needed to have
a domain where
he could work
and do his
algorithms,
his experimentation
and he had a
space that was
in a very
urbanized
environment on
a very harsh
street edge.

So we created
a cabin for
him in the
city in a
shop.

So essentially
one had, I
don't know whether
you can see my
cursor, but there
was a sort of

street on the
bottom edge of
this image.

This is the
plan of how we
built it.

There was a
street on the
bottom edge with
a lot of passing
traffic and
people.

And in order to
kind of recalibrate
the space so that
it worked to
generate a
positive
regenerative
environment for
him, we
entered from
the back and
we arranged a
sequence of
entry spaces that
were essentially
black box
experiences that
would kind of
launder the urban
fog or the urban
pain of the city,
the hard fascination
of the city, launder
that out of his
body before coming
into this timber

domain where he
would rebuild and
connect into the
truth.

That was the
overall kind of
existential
phenomenological
journey that we
were setting.

So as we were
walking into the
space, we were
walking through
these deep kind
of lost spaces
that were all
black with a
scrunchy kind
of floor covering
different acoustics
and then you would
go through this
door towards the
light.

It would open up
to you and you'd
be bathed in this
kind of warm
light where each
of the timber
elements was
retained its
organic crusty,
you know,
wabi-sabi-esque
character.

And, you know,

it worked so well
for him.

He's never kind
of, I mean,
he's always been
extremely supportive
of the fact that I
did this for him
because it was kind
of, it proved to me
that on a one-to-one
level, if you listen
to someone's needs,
you can then create
an environment that
has a major effect
on them psychologically.

Okay.

And then part of the
journey, and this is
part of the kind of
rational trajectory
of the work, we
worked with a very
preeminent company
called BHP

Billiton.

It's the world's
largest mining
company to do
their headquarters
in Melbourne.

And in this
project, we're
able to change
the relationship
between how a
corporate office

worked in a city
and how the
relationship between,
in a sense, an
institutionalised
corporate entity
and the footpath,
the streetscape,
and the history
could co-evolve
so that what's
typically a very
dry and
alienating
environment could
somehow evoke the
past, the memory
of a place, and
create a different
material reality
and material
associations for
everyone who uses
the building.

And this project
sat on an old
site that used to
be a theatre, and
it sat in a
district of the
city that used to
be a manufacturing
district.

So they used to
make fancy goods,
like they used to
make jewellery and
watches and things

like this.

So what we wanted
the building to be
is some sort of
jewel-like project
that spoke about
that and sort of
heroicised that
history.

We brought that
into the spatial
construct that
linked the two
streets that
bounded the site,
and we created

this kind of
sequential journey
that detuned
people from the
busyness into the
quietness
through the use of
natural materials.
structures.

And you can see here
at the back of the
site how the
warehouse district
gets mapped into
the structure and
embedded in a
slightly different
way, using cut
glass to make it
more ceremonial and
exotic.

And that's how the
project kind of sits
in the city.

And you can see the
echoes of the
relationship between
the Gothic.

there's a river
also sits down here.

So there's somehow a
relationship between
the environment,
the Gothic, the
river, and the
past, all sort of
assembled into an
architectural approach
that's specific to
that site.

In terms of
residential work,
we also have been
working on some
very interesting
edge conditions in
the city of
Melbourne, where
we've been able to
explore the tall
building typology and
how tall buildings can
work as civic
buildings.

And this is an
edge of the city
where lots of
parades have happened
in the past, edged

by parkland, old
buildings that were
built in the sort of
mid-1800s using
stone.

And so what we were
keen on doing here is
establish a new
architecture that's
new architecture that's
away from the glass
curtain wall driven
way of making things
and return the city
back to heavy
materiality and
extruding what would
normally be a palazzo
scale, mid-scale
building into a taller
building.

building.

So essentially your
physical experience of the
public realm was about
continuity.

But then also the
expression of the building
on the skyline because
cities are seen from
afar.

Our city here is a big
city in a very flat, it's
one of the world's largest
basal plains.

buildings.

So it sort of erupts out of
that basal plain.

So buildings can be seen
from afar.

We wanted to signal what
the city was about from a
distance using a different
approach.

And then connected into this
sort of edge condition of
the city.

So you're very much taking
part in the activation of
streets and the definition
of street edges and seeing
architecture more as a kind
of passive tool for containing
space and defining edge.

But then also creating nooks,
in this case kind of cave-like
nooks around the edge of the
building where you feel the
weight of the city around you.

This is a photograph of how
that edge condition works.

So embodiment, I'll perhaps
go back to this.

Embodiment is more about, and I
guess what I'm trying to
establish here is sometimes
you're trying to rationalize
from memory and from a kind
of fact-based, almost like a
simple trajectory to create a
building.

Other times you're trying to
capture an emotion that's
perhaps more complex and
evasive.

So I call this process more

like embodiment.

And this is a process that we use for the new Australian embassy, which is just being finished in Washington, D.C., just about to be opened, where we were looking at how a material like copper can establish itself as the material to denote a country.

And the experiences of that material in the city of Washington, which has a very sort of ceremonial and hard to grasp institutional reality, and to kind of insert something that was much more open, deinstitutionalized, welcoming, and also textured in such a way that it drew or created meaning for people outside of the building, but also as you come into the building, what those experiences are, were about Australia.

So we're looking very much in drawing experiences in plan form, and then using the copper to evoke this mysterious ecology of Australia that's particularly evident in the ephemeral salt lakes that you see in the desert regions.

So if you ever fly over Australia, it's just the most beguiling experience that you'll never forget.

And it was trying to establish a dialogue with that for people in Washington.

That's a picture on top of Uluru.

And you get to see the kind of monumental, modular monumentality of that.

And we wanted to establish something that was kind of rock-like or big and bulky

and solid, and then insert these fishes into it.

And these are some of those sketches.

So this is again, it's interesting kind of working remote from another city and inserting a piece of a country's DNA into another city.

But here we're kind of looking at the dimension, the dimensional reality of the building and how texture can get worked into a building from outside to inside and how nature can establish itself in different ways, different experiences.

So breaking down the form of the building.

And then we started to look at the because we had to what machines and technology would be able to produce the material that we wanted.

So we actually get into how the heating of the copper would work and how we would create what kind of organic replication.

So a process of making material that has a lot of idiosyncrasy associated with it, but that's replicatable.

So we use this heating machine, a top down heating machine, which the material will be rolled through and establish this polychromatic kind of burnt finish within which we had a just an infinite amount of variegation.

Very beautiful.

So this whole process evolved around the making of the building around using that polychromatic finish to create an echo of Australia on the outside of the building that would stay silent, but beguiling.

And then the way that those different panels were used was to create a kind of inflection.

And that inflection that we had on the facade was coming

through from the rhythms of the forests that interspersed in the deserts.

And that was kind of mapped onto these sequences onto the building, which then became kind of fractionalized and more granular the closer you got to the building.

And we would study that in virtual reality now because we're able to, and we're able to add the environment around it.

So you're able to choose your time of day, all those finishes, and you have a much more closed loop reality when you're designing now that can get very much close to the finished position, the finished state in advance.

So we're able to have a much more exacting feedback loop design system to embed those feelings and emotions into a structure and get it right.

So this is the kind of flickering and ambiguous layering that I was trying to talk about earlier in an actual forest, eucalypt forest in the high country.

And then that would come into the workplace environment through a sequence of, there's two atriums that are co-located.

One's a public atrium, one's a private atrium.

The private atrium is a hub for all of the different disciplines in an embassy to come together.

And so what we're trying to create here is, it's all made out of timber, and that timber is being used to infuse the environment with an association with Australia, in particular, the light, the striated light, and the raw timber working together.

And the spatial sequence being kind of open and expansive and tall like a eucalypt tree.

So eucalypt tree is super tall, has a big open canopy, and it has this small leaf kind of delicate light that comes through it.

So all of those feelings we're trying to trap into a workplace setting that we know both symbolises Australia, but also is healthy for people inside the building.

So this is about healthy workplace philosophy being used as a ceremonial gateway to Australia.

And these are the further images of that.

And that's as it currently stands, just about to be opened.

And that's the copper standing five blocks north of the White House,

at the sort of edifice of Australia in Washington.

So as I guess we were developing a human focused approach, we also, you can't escape your responsibilities to the environment, planetarily speaking.

So we also worked with aspects of the circular economy and looking at how a temporary building can utilise wood in the form of paper.

So we created an artwork stroke building, a temporary museum for the National Gallery of Victoria in a competition entry, which looked at making a timber modular gallery with paper rooms running through it.

So the paper, which is obviously traditionally been used as an element that you would draw on or paint on. But in this case, the paper is used as a building material.

So the building material itself is, or the space itself is the canvas.

And because of its translucent nature, we were looking at how the environment around the building would map on to those spaces.

So in a sense, what you're walking into is a spatial, immersive spatial experience of the environment moving around the room.

Through the, you know, the sun's movements mapping different realities, whether it be trees or the framing of the building into that room, and for that environment to change over time.

So with the rain and the sun, the actual degradation of that environment would also be present in the environment.

And then also you were able to infuse the, we used to use primitive dyes in Australia, the glue that comes out of eucalypt trees to combine with ash to create a kind of cement.

So the idea of using those different natural glues to create smells, eucalypt smell, for instance, is used as an antiseptic.

You can use a paper material, like a kind of sponge, that can emit smells and almost create a kind of natural outdoor environment inside.

You know what I mean? So we, we were interested in developing those naturalized experiences in cubic spaces.

And then basically collapsing the whole building and recycling it and making it into a book.

That was the whole concept. So it was a pop-up building that was made into eventually a book.

We've also been working in, in Asia. This is a project we did in South Korea two years ago.

And this is a, an embodiment project where we essentially understood, it is in Seoul, that the city of Seoul is like 80% of South Korea is made out of granite mountains.

So it's a very mountainous country. But Seoul is in a kind of delta.

So, but it's surrounded by these granite monolithic hills.

And the hills are used to, to, for people to escape the city.

So they go up the hill, the mountain, and they see a different future.

They see a different life. They get connected to nature and then they go back into the urban realm.

So what this project was about is encapsulating that natural getaway type space, but bringing it into the city to change the dialogue about what a city could be in the future.

So we, we would, let's say, decode those sorts of things and apply it to personality types, whether it be a company personality type or individual.

We decoded process of how those mountains came into being and how the geometry of those kind of geode like, I think they're called tor structures, how they would have that formed.

And how they've been ceremonialized and celebrated through Korean art, through Asian art, you know, the whole idiosyncrasy of a piece of, a beautiful piece of stone or a piece of wood.

And also, it's, I didn't know this at the time, but South Korea is also the world center for these Neolithic structures, which in fact, were hollowed out and used as primitive buildings, primitive, ceremonial places, let's call them.

So this relationship between the landscape and the identity, the cultural identity of South Korea, as seen through this geological construct being made into an office building that provided this health benefit.

So how did those health experiences work on a hillside walking up and how could we translate those experiences into a building?

So we, we, having absorbed all of that, we would then go into sketch mode and track that geometry and communicate that geometry into plans.

And into elevations and into elevations and then into elevations and then into experiences of how you would approach a building.

What's it look like at night as well as daytime?

How does the ground plane, car parking, roadway work with the subterranean retail?

How does the, how do all these relationships work if you follow that, that geometry?

[illegible]

we're trying to say here that an office building isn't focused necessarily about work and it's certainly not home it's something else and the something else thing is this sort of natural experience or place of elation place of joy where you go to to get enlightened so the idea of the building being a kind of tool for enlightenment to generate better ideas more progressive ideas and using nature to achieve that those links so that's what all this is about it's about sort of well in some sense it's sentimental but it's it's true having gone through the process of embodying those physical experiences on site trips that we did and photographing that and studying the cultural identity it's then allowing us to then create that matrix of experiences into a form which go top to bottom across the building and form a different dialogue with the public realm

so very much a building that changes throughout the day
and forms different dialogue now in terms of formulas and i think formulas are very relevant topic
the first formula that we put together was for the new royal children's hospital here in melbourne 2007
which was opened by the queen where instead of um let's say
forming a kind of um
that's a sort of how to call it an artificial uh accord with the children who were being um
um dealt with within the within the environment of the world children's hospital we saw them really as
small adults as opposed to children so we saw them as sophisticated uh people just like adults so so
so the first issue was to not create a language for the building that was going to be somehow derogatory
or form a poor a poor support structure that's a poor support scaffold for the children so we wanted a
sophisticated structure that could again elaborate that internal world of the child
think about how illness works in children who are stuck in their beds for say a month bed bound
and institutionalized and that sort of fearful
relationship between doctor patient patient in big billion dollar hospital change all of those
relationships so that it's no longer an institution but a place where you again not at home
you're not in a hospital you're somewhere else it's you're in a place of enlightenment and it's that
enlightenment
which improves your health and restores your energies so seeing children differently than what we're told to
think of children as being these sophisticated creatures that are curious about the world and
assembling cognitive cognitively assembling their world every day just like we are and forming identities
in the same way so we sort of created a what might be called the first health code solution
and we didn't just use the solution genic concepts that came from aaron antonofsky which are all about
coherence we bought in um concepts of evidence-based design which stemmed from roger ulrich's work on
natural light and then looking onto nature the kaplan's work on soft destination and nature
and their concepts of attention restoration theory and we we looked into Reggio Emilia schools in Italy
and we perhaps added another two or three concepts on top of that and we use the assemblage of blended
formulas
to create hospital so it's not strictly about either evidence-based design or solutogenics it's about an
assembly or diversity um but essentially we're creating an environment that was made into an adaptable
building but the environment was the was the driver and the environment was there to form an escape from
habitual activities and that would form a kind of soft interface a soft and reflective interface
that would retain your attention effortlessly so this is all about kind of
capturing a quality in a building that is very similar to the qualities that you would have walking through a park
where you're continually drawn into detailed areas and where you see the movement of light and air and
the general ecology as a living environment surrounding you um and washing through you so the idea of
having
a kind of soft washing of a soft environment how do you go about making a hospital out of that so we
systematize that this is just a diagram of how the solutogenic process works in terms of building coherence

from

the the focus on building intimate spaces as well as public spaces as well as family spaces changing the relationship between how care goes from being you going to the care to the care going to you so the model of care here is about doctors and nurses in the healthcare system going to the child not the other way around um so it's complex in of itself and that was a major tool for how we designed the building but then we're always cognizant of the natural nourishment that we get from natural light natural air currents pointing the for instance the building towards the sun so we embedded the circadian rhythm into the building and essentially made a building that was very similar to forest therapy um by mapping the journey through a park and translating that into a building we would naturally have a good environment for people to be in so so this is the first time that we're really starting to focus on the pathways between the mind and the environment and how it changes our hormonal uh structures our how it affects our central nervous system etc to produce positivistic results so this kind of linking between the environment and the body the mind and the body and architecture for the first time was established for us so in terms of intuition the idea was to we had an urban edge condition but a fabulous park that we designed a building into so we what we were seeking to do is to push the building and migrate the building into the park migrate urban form to point it towards the sun so that the the building ended up pointing due north and by doing that we're able to create um parkland around the majority of the building um and then what we were doing is embedding through drawing experiences of that parkland setting onto the outside of the building and through experiences of entering and walking through the building and then focusing in on the child in the bed and what their own experiences of self-destination could be within that environment to make sure they're restored as best as possible so we created effectively a model of salutogenesis into a building this is a 1.2 billion dollar project it had two and a half thousand rooms inside this building and we were dealing with all ages because it had researchers you know doctors nurses researchers that all ages focused around children who would start their journey within the royal children's hospital as babies and perhaps still be going there as adults so we had to deal with all of the age groups in the building this is just a focus on how the night and daytime use of the building works in relation to nature the celebration the historical cultural celebration of nature in it's kind of picturesque australian naturally picturesque um it's like a country garden and using those features in the building this is a picture of that picturesque landscape in victoria this is our site uh before it was built on it used to be a cricket pitch and the cricket pitch it points due north interestingly points due north and that became where we put our streets and we created a street that would form the at the central social connection device through the building um so this was us mapping uh qualities visceral uh textural uh warm and inviting qualities of the park before we designed so we're studying the park studying how how the layers of soft fascination work how shadows work what's the sort of geometry of a tree what's the that's a strange ergonomic quality of the trees how they cluster together what sort of light comes through a tree how does a bark texture work and what's the tone or quality of that tree so all of this was documented before we started to design and then we were embedding those experiences into

circumstances uh we developed up a color palette out of that we created um walls that that reflected the kind of layered soft fascination light quality of the park we went into great detail this is almost like a one-to-one study of an edge of glass and how dots and patterns and soft curves were being used to evoke that

and then we created a bunch of experiences throughout all of the two and a half thousand rooms that evoke that

those experiences into architecture principally on the basis that the relationship between you as a person looking out into nature was a healthy one and nature was brought to life through the sun and what we're essentially doing is trapping the sun nature energy and bringing that into the building that was the essential relationship and then we were personalizing it so avoiding institutionalized replication we were interested in having each of the two and a half thousand rooms different to one another as well which is pretty high order of uh complexity uh this is the plan of the inpatient unit which was shaped as an organic star shape uh for practical functional reasons but because we did that we're able to break the experiences of the nurse station relationships with the children into smaller packets of space

uh so and then we were able to open up the corridors into collective space rather than corridor space to make more of a home and a kind of focus point for for the different spaces so i guess what i'm trying to say without going into great detail is that within each of the spaces either at a very microscopic detailed material matter perspective or spatially speaking how rooms relate to corridors or how building relates relates to a park all of those the different scales they were all treated with the same level of care um so again focuses on um again associated with the soft vaccination how this kind of dot matrix that was encircling the building how it resonated with the eyes and how it would change with time of day that would come to life we used energy saving techniques to surround the building with these glazed sun canopies that were that were made out of colored glass made out of dots of different colors and different shapes and because we're able to map the trees outside into this space the and because we're making these sun canopies out of glass we're able to create a kind of transcendent experience of the edge of the building that was bringing that energy deep into the heart of where the children were living and this is how the external building came about so we're literally translating the foreground of nature and wrapping it into the building to form a blurred relationship between building and landscape so it's very firmly embedded into the landscape it's an extrapolation of the landscape

so in terms of the geometry um it started with the tree geometry and you can see this in the shadows but then if you look at those kind of arms that are coming out of the tree we wanted to utilize that geometry for um how we constructed the larger spaces whether it be waiting spaces or the bridges that that or

the textures uh in the internal cladding of how these um yeah bridges that that came in and out of the different functional areas of the building would crisscross the street and how they would assemble themselves into these kind of nooks or torsion sculptures that were very similar to the outside world the scale of a bridge or the scale of a staircase how those large scale pieces would relate to a child's

mind's eye in even though you're in a large space these large scale pieces were being assembled to kind of de-institutionalize but also form a sympathetic relationship to a to to the scale of a child and the joy of a child's mind's eye and on the outside just looking at how these sun shades again would replicate um pattern language of the trees into a kind of foliage that changed color and patina depending on the time of day so that was kind of very interesting process and it after i guess we finished it it um it changed my paradigm or changed my thinking of of architecture as being a tool for health full stop so since then since 2007 all of the work that i've been doing has been associated with health and seeing cities as generators for health um and so once you once you get to that position you then start to think well is nature or our job in translating nature all associated with the natural world like a royal park and using soft first nature or is it something that's more associated with human psychology and the nature of being human and i guess the latter concept uh came into being kind of more recently with uh work that we're again doing in south korea which we called trans buildings which are less simple translations of nature which actually focus more on human nature in buildings and this project was in itaewon and it was a project for a private members club and you would probably know soho house and buildings of such life and big cities where members come together in private realms to celebrate big members um what we were doing here is changing that paradigm and forming a private members club that was much more public um so unpacking a singular building and making into a kind of civic domain so creating public space out of a private building and then in a in the site forming a different relationship between buildings in their environment um making a more ambiguous relationship um so the buildings were seen as sculptures and kind of like non-buildings um they were they were broken down into pieces so it became a district as opposed to a building and what we were trying to do again is to change the psychological relationship between the people inside the building and the building itself but also the members of the public who are looking onto the building and forming a questioning relationship so kind of imbibing a an existential crisis if you know what i mean so seeing a building as being a partner in the journey of existence in the crisis of existence and the questioning of existence um and not necessarily being a utopia but being kind of just a living piece of city and we're also interested in how the new world of virtual reality is coming into being so trying to establish a building that can go into the metaverse as well so we co-located the idea of a new virtual building with an existing building to form a more hyperreal um form of architecture which i'll come on to perhaps later in the conversation but the environment the environment was a very interesting uh context where we had the most international district of soul a place uh fraught with social uh friction um a disheveled environment that was very popular came to life at night so a lot of kind of non-likable features in in very broad sense um that naturally come about in cities and embracing that rather than critiquing it but trying to form very beautiful cinematic environments which were highly evocative making buildings that were charged with that kind of psychological tension using materials that were hard to describe so in this case it was 3d printed wood and seeing the building as an outside face but having an inter a completely different internal world

so sort of in a sense evoking what we're like as people we have an external appearance but we have a highly complex and changing internal world forming similar relationships in a building and then making a building that was ambiguous as to what it looked like having gone through um an interest in in this essay by jg ballard his his fabulous uh written piece on the garden of time we then started to we don't have time to go into that but we then got into translating all of that into art pieces which then came into physical experiences of the buildings

um disappearing underground and the exploration of an underground world so essentially arriving at an assemblage of buildings half of which are above ground half which were below ground appearing as these concrete simple prismatic volumes

and quite stern and monolithic

the walls of the buildings and using water to at the base of the buildings to cleanse the buildings and the environment to then looking at how the internal world worked as a more organic place and contra to the prismatic volumes on the outside looking at a much more bodily like sequence of spaces on the inside so whether you're arriving at the retail or the rooms within the private club or arriving by car you're sort of arriving inside a body and having a very different relationship and then the two worlds then connected and we started to look at how windows worked in the building how the concrete would reduce to

fabric would would transcend and change into glass and how different relationships were working through different spaces and how that would become organically expressed into different sorts of thresholds which were more more bodily type thresholds to form a building that was kind of half building half body and then looking at how the concrete in the evening would then recede back to become a more translucent material

so the idea of the buildings kind of being more translucent during the night time and solid during the daytime and having that ephemeral quality

and now lastly we've been working on the new hydrogen economy in south australia in a place called wyella looking at how to use new materials like green steel how to repurpose industrial buildings but looking at how principally the ecosystem how the industrial ecosystem can create a more humanized uh urban world so we've got we're on the edge here of looking at how the urban reality of this new city that's going to grow because of hydrogen energy how the relationship between energy the earth manufacturing and urbanization triangulate into a more holistic paradigm by looking at all of the different ecosystems being embedded into one another

the other so starting with nature and the natural reality of the site the ecology of the site

um how that the ecology has been manifest into structures whether it be timber old timber piers or old existing buildings into a kind of embedded memory of of the site

so we're recording in a sense what the memory the cultural ecological

memory is of the site and then we're translating that into a shape language an experiential language that then overlays into the new housing project that we're doing to generate experiences that capture those that that sort of existing experience and manifest into a bolder bigger identity for the city so

these are sketches as they translate into new house so sort of balance between the sculptural world of found objects on a beach made into a piece of architecture forming a new city out of all of those embedded memories

so i so i guess daniel and the audience if we've got any audience left all of that what what all of that constitutes is a kind of flow and the reason why i presented you like that is it represents a kind of intellectual theoretical practical artistic flow that represents how creative cognition works and i guess the overall ambition is that because of that coiling process we end up with a much more coherent and friendly human celebratory world and that's a world that we all need and deserve

wow i guess i would have said i'm not sure where the spiral ends but of course it doesn't end yeah it doesn't end absolutely yeah well please continue uh well i i mean from my perspective i've i see the institutionalization of the world of architecture the world of property development the world of law the world of politics the world of industry um having behaved poorly let's say in the past or perhaps narcissistically speaking um evolving systems that serve their own ends in their own ways but we're now sort of in an environment where we can actually bring all of those pieces together and to form a self-reinforcing self-reinforcing you know networks and thoughts and processes that can actually inform and and manifest a better physical world so it does mean that as a artist architect you are interested in politics you are interested in industry you are interested in ai for instance um because you're interested in consilience you're interested in tying together the different disciplines just like and how nature works you're in dialogue with the environment so in a sense all of this is is naive in the sense that all it is is a replication of natural process

it was an incredible presentation and i have many pages of notes but first to kind of lead into that i would love to hear about your engagements and encounters with carl friston et al

um well like here um i went through a very similar um kind of presentation and i guess what was interesting i didn't know how it would resonate or whether it would whether it was just me fantasizing that it had something to do with free energy and active inference but i had a very strong hunch that it did and i guess what i'm what i got from it that very very much indeed does correlate very strongly with um with the embedded principles that carl's work is all about and um i think he was very pleased uh to see it being translated in this discipline in this way um and i felt relieved to be quite to be quite honest you daniel talking to someone of the stature and caliber of someone someone like carl um to be able to talk to be able to talk directly with him and the people around him who are all excellent um was was was extremely fascinating and beautiful um i mean i see it as a kind of the the way i i just see it as the way forward um and i'm excited to understand how it can translate into into a process in a way what not in artificializing that process naturalizing it but but able to encapsulate the complexity logically so that people can understand can get on board so we can de-institutionalize this this logic and spread the word if you know what i mean um spread it around and encourage participation and experimentation um so i got i got the kind of if if you know what i mean the collapse of boundaries in the car in the conversation with carl because because it can go into all areas of course it goes into all areas i'm just showing how it translates in the way i do it but um

um i think it needs a long time to kind of just sit there and and discussions to just to to spring from here you know to spring to get going um i see this spinning into all sorts of different areas so i mean at the moment we're looking at how to utilize this in coding terms so i'm interested in computation and statistics i'm interested in urbanism like making new cities um so how we translate this into we're building a 200 billion dollar new rail loop around the city of melbourne how this can translate into new districts and new cities and how we can break down this typical artificial boundary between new things

and i don't know new cities which are typically unfriendly unloved and unwanted into being coveted people aspiring to move move into these things right and we never got that right ever um i mean since paris you know we've never got it right so the idea is to actually is to get something finally up and running which represents an opportunity to to do something at a regional level a local cultural community level that perpetrates evolution at different scales in different areas through a process kind of open source

very fascinating again there's there's so many places that i want to just touch on because i i think you do present so many contact points and ways of expressing that will resonate with active inference enthusiasts

and yeah i i guess the starting point is the learn dwell nook so yeah you were taking those measurements because they were informative you were updating your cognitive model from your sensory input and that changed your generative model it also may have been actionable information it may have been folded through an

and iterated realized process back into the design and then fundamentally that's an attracting space it's an attractor or dwell

space just like if you had a landscape and you were shaking up a ball on the landscape it would dwell as you turn down the temperature and slow down the shaking it would dwell probabilistically according to the features of that landscape

and that concept of attractor points attracting sets and then also wandering sets and moving nooks which is something that may be able to happen in a digital space or it might be a mobile element or movement is all just a question of over what time frame that idea of the attracting set and of things existing by virtue of what they're attracting set is that's the heart of the free energy principle

so starting with the dwell on a surface and then i'm sure there must be a point in the design where it's like so we saw 50 this and 50 that do we want to carry that forward or are we looking for like 80 20 or do we want 40 40 40 20 with a new affordance being introduced through landscape modification it completely fluently um touches on it and i just love that learned well photography because i think probably many of us with cameras in hand now have our own little peculiar brand of photography and then what does that say about us

the world that that sits in terms of free energy active inference and how cognition works in relation to creative process creative cognitive process as let's say creative cognitive process being kind of symbolic representation of cognitive process as a sort of exploration as a projection what do you think about that well we may have to revisit it multiple times but here's what i wrote down perhaps touching on it you

showed a image of of a photograph of some metal that reflected the idealized and the actualized construction of the eye beam coming together yes and that both of them had differences that were uh materialized differences through time and weathering and context and it was an assemblage so there was like all of this richness and i thought wow it's an eye beam that's like the self that's like the eye the eye and then there's a comparison of the idealized eye even of just approximately idealized eye like i could be standing up right now i could pick up that object

that kind of idealized counterfactual adjacent possible the abductive the associative the creative that exists in our liminal proximity of the eye and it's like yeah and then it's all about that what the image that was shown with this sometimes horrific but sometimes beautiful juxtaposition of the real and the actual eye or moving in a little bit out of psychology like the real and the actual self-model and then our calibrator on that is variational free energy yeah

but you know this kind of how do you say it um

uh the concept of the the talk was about naive architecture or naive approach which is a kind kind of you as vessel for that complexity so you're sort of living manifestation of that complexity as a person as a human being you're not you know i'm not a tree i'm not a plant i'm a human being but the cognitive process and natural biology and the coincidence of those two realities is being brought together as a cognitive process that manifests that expression of the well it's the it's sort of the first loop is you absorbing and then expressing and then the expression becomes the physical manifestation of that thought process being the substrate that you live in right so that whole churchillian concept of you know buildings make you know we make buildings building make us that whole loop

system um you know it's obviously been been proven to be to be right and cities can either be pathogenic or solution genic you know how many cities in today's world in modern sense are solution genic cities none you know you go you'd have to go back to the villages and towns and blah blah which which are very much tight manifestations of humanity in in a sense in a physical manifestation sense um and the abstracted i would call it excel spreadsheet logic where you take one element and then extrapolate cities out of that one element loses all of that nuance so now being in a situation where we understand that nuance and we understand the importance of that nuance there are elements but also links in that nuance matrix that we can almost calibrate and we cannot we that we can capture them we can embody them we can recreate them them and now we've got this kind of opportunity within that sort of febrile matrix of stuff that's in part facts that are

math-based facts in part just visual uh reflections or actual photographs um

um but we're now able to kind of like bring all of that together as a sort of more complex arrangement of things to manifest something as the new paradigm um that forms that sort of partner in in our cognitive process uh that's not an enemy to our cognitive process that forms the coherence and the um natural it's our natural place if you know what i mean it's the sort of heidegger space it's the place of dwelling yeah a lot of angles it's the william blake dialectic songs of innocence songs of experience

but i want to connect again the visual representation what what do these visual what what even are we looking at it's something i asked myself probably on two-thirds of the slides i was like is this if there wasn't a person in the frame i was like yeah is this a microscope is it vr is this aerial footage and sometimes probably even if then later on oh that's four trees but it's like but is it and all of that richness of of the the temporally thick interpretive process and also the relationship between like the synchronous audio visual and the eye saccades a lot of other features that we can quantify um there was a scale freeness or a scale ambiguity a system ambiguity and the mathematics of active inference in the free energy principle are scale free or we could say scale friendly which is to say they don't host or hinge on any specific or particular system or scale it's not a framework about brains but it can be used to paint or express models of brains and so that meta theoretical position which may have more resonance with the creative encounter or the kind of like massively collaborative and global projects that you described that setting from a praxis angle especially with augmented and synthetic intelligence may come to be closer to the process of active inference modeling than 1995 or 2015 writing a statistics program yeah i think so i i'm with you on that i think it is i think it's really it's kind of experimental actual experimentation of of the free energy the world of free energy and the successes of it we can see an experience um that's the joy of it and i think i also would like to say that with studying how creative process works you're almost peering inside free energy as well i mean i this is just me talking and i don't know speaking whether i'm right on this or not but it just feels to me that it's a rich a rich source of um information to unpack that actually could provide some really good strong direction for the future the the piece that really left an impression on me that makes me think you you said right there the successes the realized successes successes only in principle or hypothetical the real ones are real and you mentioned in the copper for the washington dc building that you wanted to make it idiosyncratic yet reproducible and that's like the magic quadrant like yeah we have technologies of reproducibility without idiosyncratic plus externality plus affective um pathology pathology etc and so this quadrant or space with the idiosyncratic and reproducible and then it was like it's a top-down heating machine and so in some ways the realized material artifact becomes like the crucible or the eye of the needle where the generative process the process that leads to the artifacts deposition um or accretion or however it's built it's a demonstration of the sense making and the aesthetics and the technical capacity it's like wow they brought that rock from here to here and then yeah did they do it this way or that way and people today discuss and debate how did they do that building how did they do that building and so it's like the archetypal logistics operational setting construction physical construction is the tip of the iceberg of the physical and informational supply chain and all of those for those who have the generative models to see we look at the obelisk from rome and that is like a big rock and it ripples for those who have knowledge of that setting but also the footprints of an animal to another animal or the pheromone of one ant nest mate to another ant nest mate or all these other like infinite last miles and those are all yes ands with each other they're not in contradiction to each other and neither is it a contradiction to have a unified way to discuss and approach that yeah so so within your work and within the studies of active inference how are

you critiquing or unpacking modernity in cities in buildings how is that being how is the world of active inference being explored in the physical domain or is it or is it being explored or but when it is being explored how sophisticated or long term has the research been all great questions the first thing that came to the world of active in the world of active in the world of active in the world of active and the physiological and the interoceptive and while there has been for example models of movement within a maze is to really to really get to the point of a simulated model

is to be more than just an intuition pump which can be more than enough itself but for it to be more than that really at this point in time requires requires a team professional effort that to the few extents it is happening is probably happening in proprietary settings but we do everything we can to highlight the open source capacities and the kinds of models that people are building i think one setting that is pointed to a lot but also has not received that much empirical or open source work is like ecosystems of shared intelligence instead of just talking about the physicality of the internet of things or iot or everything in that space being able to integrate the spatial and ergonomic like you pointed to earlier with efforts to provide something like a meter stick but we're not just looking for distance we want something to calibrate the construction or the architecture process and at hand was the industrial or potentially modern system and then what i saw right from the beginning of the presentation was like a yearning for a compliment or a juxtapose or the shadow of that and affective cognitive science now being able to kind of square that circle or however you want to think about that where it's like well we could organize our project around people's sense of narrative fulfillment and that could have disastrous ecological consequence so let's care about the narrative and the ecological okay but then the business might not exist all right bring it into the generative model what else what do people think about what we have now and that must iterate that must be intermodal and so that allows the imbuing of all these different aspects that that for me were evoked when you talked about like soft attention whereas in science and technology and math which was only one third of your three piece pie a soft attention can be a boon for a beginner in scoping out the space it can also be um a pleasure and a tool for an expert who's able to have a soft background and then sharpen attention um but to be useful one has to be on the other side of that innocence and experience and that's a tough learning curve because there's and no one is ever going to be able to have all of the hard attentional regimes there's just too many it's just like trying to know every language or like no it's like it doesn't even make sense it's not even preferable like for any number of reasons and then it's like well so what do we do but it's like but there we are scratching our heads talking about how we're going to create livingry and make it work physiologically if not far far more than physiologically

as a group as a team as probably discussions have always happened leading through to the enculturated and lovable structures that we love

and so i love i love how you use words it's fantastic um um you see i would sort of say that diversity that what is enabled to get hold of to grab in and bring in to a creative process if you have a for us or agile process allows you to bring anything in think always made me feel as though when a project came about i would always say well there's a million buildings here which one do we want to build which one do we want to design and that was kind of a fairly alien concept to a lot of people in my industry or in my education certainly people wouldn't want talking those terms because it had always been trapped by some cultural doctrine that one had to be a post-modernist one had to be a deconstructivist a regionalist a modernist this or that there was always some sort of virtuous um theoretical underpinning to something that was determining what you were basically institutionalizing what things should be like for whatever reason for power influence money whatever it is we were trapping a process a thing um codifying it and then telling everyone that they had to be in that box whereas what we're talking about now is totally liberated it's totally free for being what it it should be so my yearning for craft my yearning for vernacular architecture my yearning for that hard to describe soft granular environment that i like so much in the old world in inverted commas or or seen in agrarian structures um i can now talk about in terms of how professionals talk to one another how teams are made how um

the intangible has now become sort of tangible through the ability of talking about it in terms of free energy um neuroscience it's enabled us to actually bridge the gap between qualitative and quantitative and that's the joy of it of being able to go from an artist wandering in the bulrushes feeling utopian but creative and curious manifesting things to really having a quantifiable opinion about something and being able to capture it compartmentalizing it without losing the nuances compartmentalizing it and explaining it to someone and then you know it'd be like sort of engaging

someone to design your house quite often what would happen you'd end up with arguments with your architect

you'd end something end up with something that you didn't want um you probably went through some sort of mood board analysis or gave him some sort of brief or them some sort of brief um but there's always been this hard boundary between what you want as a person and what someone else is giving to you you know and i think what this is enabling is a soft um um sort of navigation or translation or um arbitrage this sort of soft and giving arbitrage between you and the environment you and someone else you and a large corporation that enables both parties to come together around a common set of values that enables that negotiation of the arbitrage to work in everyone's interests so i think what sits underneath all of this it's translating yes into an architectural place making philosophy and place being a kind of maybe closely associated with coherence i would say but anyway the idea of it being a negotiation tool i love that i love the environment as being an expression of fruitful coherent arbitrage of relationships awesome and synchrony generalized synchrony with the environment not lockstep with the

environment but synchrony like turn taking and meaningful dialogue and alignment with our environment our built environment information architectures our social architectures and and i think this is something that hopefully we'll see unfold in so many different ways is like how do we hold the moment like the aesthetic moment the lived um the humble home environment moment or whatever moment the affective moment

outside or or even before we even quantify the affective or the cognitive like the real real primal moment and it's like maybe one day we'll have tools soft enough to touch i don't think we have them today so what is one thing that we can do and maybe not even regret ever looking back something like create a buffer or a container for the vital moments and so it's like let's let's just make sure everyone's at the workshop on time and that this is the way that they like it and and all these features and be very thoughtful and mindful and value aligned up to the moment of the craft and then if there's more to say it may be right to say it and impose then and constrain then but if there's nothing more to say after you've like brought everyone to the dance then that would also be fine and so then the challenge is like but you have to make a building like you're being paid for a building and then there has to be that received public or civic experience that is not like oh we'll watch this to our live stream and check out this or that analytical framework there has to be the real direct and and i mean those are the kinds of knots and tangles that it's just amazing to hear your your journey and your way around they're kind of i call them a call i've just written this down actually

nook machine i think what what i do is like nook machines and so

i think by helping the world by creating nook machines we end up generating a more truthful honest coherent amplified truthful kind of expression of how we want to live with all of the different instincts activities that we do every day right and it's kind of like the enabling of all of those activities whether we want personalized in you know intimate moments or expressive outward moments but the health of a human being being that sort of inner and outer world being being enabled to effortlessly assemble every day melodies and that sort of idea of a building let's say an environment capturing a multitude of different feelings not a singular feeling but a multitude of layered feelings to create melodies and so often what happens in symbolic architecture is you're really drilling down in the guardrails of a singular emotive reality and you know none of us live in that singular emotive reality and we don't want to live in a single modality like that so that sort of multiplicity of modalities sets that that sort of complex melody thing and i think that's that's what we're really talking about is through that communal process or that iterative self-improving process which absorbs contrasts

and works through juxtapositions we can absorb a country's voice uh a complex singular voice into another person's voice and the beauty is that synchronicity or that moment being distilled which then enables further conversations to happen which is kind of what i think nature does to us naturally it sort of forms that dialogue um and what we're missing out of cities is in the physical environment is a lack of dialogue capacity so this nook machine thing is really about creating porosity a living porosity a living porosity that that allows people to be okay to be people you know

because we're the city in a sense is telling people that they can't be themselves and that can't be good or it can't be good for us you know

so

i think we're getting close to kind of having a kind of complex modeling process that then can feed from data that we can find let's say to improve the modeling but you know we're on a journey of course but we're at the point in time where we can probably get a lot closer right now in objective terms in rational terms to get that poetic feeling distilled and to be able to be able to tweak it to be able to sort of play with it tweak it synthesize differently sounds and the music coming from it before we build something and then we also can start measuring those qualities after after it's been made and literally feedback that information to the next project wow well in our sort of closing ish thoughts a close synonym for nook is also niche like i've heard a breakfast niche and in ecology of course we talk about different aspects of the niche and in cognitive science to the developmental and the cognitive niche and so it's like that's the map in the territory the cognitive niche is where your belief distribution is hanging out and there's a cognitive niche about the breakfast nook like we ought to have breakfast there or we're the kind of family that uses the breakfast nook for that but it's like whatever it is on the map and whatever it is on the territory the of course digital being a complexifying factor and all of this but whatever it is let's just go with that and have a path of least action on the generative model active inference with a free energy principle and then we can talk about what we expect and what we prefer like we expect everybody for breakfast at exactly

this time versus like we have loose expectations or it could be uniform across the whole day we can talk about our expectations and preferences no appeal to a reward people aren't loitering and lounging in the park for a reward function directly or per se and yet that is what the kind of quantified matrix and the reinforcement and the reward learning which is not wrong the reward is not wrong it's just a special case it's a partial constrained expression and a partial constrained regime on all sides of the blanket and so

i don't know where it all goes from there

well but daniel you know the the learning dwelling nook concept the conflation of having a synchronous um let's say density of want where you're both intrigued and curious about something but also kind of want to be in it so you're sort of learning you as you're moving through the environment you're seeking to learn and on top of that you're seeking to gain and arguably seeking to inhabit or dwell better or improve your chances of dwelling better as some sort of effective strategy for survival yeah i don't know whether that's true but it's quite actually the two components of expected free energy you said to learn and to gain epistemic value pragmatic value these are the two components of a unified imperative and we want a high reliability setting where it's like you're not here to learn we need this done that way yeah and then there's also the other end of the pool and there's all the dynamics and movements within and it's like that's that's the palette now what do we want to say yeah yeah um and what what needs to be said

comes out of the the arbitrating process uh which goes into a lot of detail because doing a building for instance takes like seven to ten years sometimes you know it's not a short process um and the reality is that they're kind of like codes it's like you're building an algorithm it's just it's a physical algorithm it's a physical embodiment of knowledge and that knowledge it creates physical experience that people intuitively experience without you know maybe dwelling on the seriousness of it or the impact of it people are at least experiencing it and so this the trouble that architecture has always had in the modern era is evaluating the power of architecture or the meaning of architecture to health and well-being and the growth of culture why why is architecture important is it important um and if it is important how do we quantify the importance of it um um and and architects and the industry tend to if they're not maybe born into environments where they naturally get it if you're born outside of architecture in a sense maybe you've been surrounded by industrial reality the whole of your entire life you may not i'm just guessing it but you may not fully comprehend at a bodily level what the meaning of that is and why it's important virtuous beneficial i don't know but but for me intuitively maybe this goes back to archetypes being an artist architect looking to looking and being curious is a foundational thing for me at a personal level personality type level but so is the idea of creating structure so the idea of creating on one hand and the idea of structure on the other so creating scaffolds and so um that's my particular personality type um and i feel as though it's almost my job you know it's like i've been born to do a job and i'm enacting that job and i'm enacting that job i'm just trying to unpack what i would naturally urge what i've been naturally urging to do my entire life to the point where um if someone says why does this space feel so good it's it's much more complex than just saying well it's got good natural light or um the acoustics are good or it functions really well there's the density of the psychological reality that people never get to talk about and there's never really been a framing of that i mean heidegger's had a good bachelor heidegger they've had a stab at it um but it hasn't had the sort of um balance um of dealing with highs and lows the different tonality of life uh the naive to the sophisticated the the poetic to the i don't know the everyday there's never really been something that's systematically drawn all of those constituent elements together and it's always become a dry dried up kind of artifact a dried up almost purposeless lacking purpose theory that sort of begins and ends and then it's just there on a bookshelf for other people to learn about but there's never really been an active philosophy that has translated and improved continuously over time and there's something about neurobiology and the fact that it bridges bridges into creativity in the physical environment that is the kind of for me it's the um it's it's everything really it's the it's it's the moment this is the moment when it actually comes together and we're able to actually use these kind of golden threads and make make some magic and that magic not being some sort of um accident but actually having a tangible realizable teachable or open sourceable kind of reality to it we're just we're just at that moment i think where we can begin that work thank you would you like to make any closing statements and then i'll make a closing statement

um i think it's the first of a lot of conversations

i just think this is the primer for me it's the primer so watch this space

is this is the bedroom is the bedroom is the bedroom is the bedroom is the bedroom is the bedroom

is the bedroom is the bedroom is the bedroom is the bedroom is the bedroom is the bedroom is the bedroom

that William Blake had on his tombstone constructed he wrote I give you the end of a

golden string only wind it into a ball it will lead you in at heaven's gate built in Jerusalem's wall

very nice thank you till number 60.2 yes till next time it's been awesome thank you thank you see ya bye
you